

Module 5 - Local RAG POC Summary

Document Loader

Loads enterprise documents from multiple formats. Converts files into plain text. Preserves metadata.

Multi-format Parsing

Supports TXT, PDF, DOCX, XLSX and PPTX. Uses dedicated parsers. Produces normalized text.

Chunking

Splits documents into manageable chunks. Recursive chunking preserves logical boundaries. Improves retrieval.

Chunk Size

Controls context size. Too small loses context. Too large reduces precision.

Chunk Overlap

Shares neighboring text. Helps across boundaries. Dataset dependent.

Embeddings

Convert text to semantic vectors. Preserve meaning. Used for similarity search.

Vector Database

Stores vectors and metadata. Enables semantic retrieval. Persists knowledge.

Indexing Pipeline

Load → Chunk → Embed → Store. Creates searchable index. Run initially or incrementally.

Retrieval

Embed query. Search Top-K. Return relevant chunks.

Retrieval Filter

Applies threshold. Removes noisy chunks. Improves grounding.

Prompt Builder

Combines context, history and question. Adds instructions. Builds final prompt.

LLM Service

Sends prompt to local LLM. Receives answer. Keeps logic isolated.

Chat Memory

Stores conversation. Supports follow-up questions. Added to prompt.

End-to-End RAG

User → Retrieval → Prompt → LLM → Answer. Grounds responses in retrieved context.

Key Parameters

- `chunk_size`
- `overlap`
- `embedding_model`
- `llm_model`
- `top_k`
- `threshold`
- `metadata`
- `context`
- `history`
- `prompt`